Video Compression Homework 1

Spatial Coding Lab Report

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1 Implementation Overview

The standard codec processes 512×512, 8-bit grayscale images as 4096 non-overlapping 8×8 blocks. The encoder applies level shifting, a separable orthonormal DCT, weighted quantization, zigzag scanning, and VALUE/EOB Huffman coding. The decoder reverses these steps with inverse zigzag, dequantization, IDCT, level restoration, rounding, and clipping. Internal zeros are encoded as `VALUE(0)`, while EOB replaces only trailing zeros.

The experiments use both Lena and Mandril. Rate is measured from the complete bitstream, including the 4-byte header and final padding:

$$\mathrm{BPP} = \frac{8 \times \text{file size in bytes}}{H \times W}, \qquad \mathrm{PSNR} = 10 \log_{10}\left(\frac{255^2}{\mathrm{MSE}}\right).$$

2 Experiment 1: Quantization Table

The provided table Q was scaled by $s \in \{0.5, 1, 2\}$ using

$$Q_s = \max(1, \mathrm{rint}(sQ)).$$

The same table was used by the encoder and decoder. The $s = 1$ case is the standard implementation.

Image	Scale	BPP	PSNR (dB)	MSE
Lena	**0.5**	**1.266205**	**37.865965**	**10.628792**
Lena	1.0	0.837616	35.808515	17.069817
Lena	2.0	0.572144	33.704943	27.706783
Mandril	**0.5**	**1.944489**	**37.499634**	**11.564236**
Mandril	1.0	1.454590	34.203934	24.699432
Mandril	2.0	1.045013	30.991261	51.755238

Table 1: Rate and distortion for scaled quantization tables.

Smaller quantization steps preserve more DCT precision, so PSNR increases, but fewer coefficients become zero and BPP also increases. Larger steps produce the opposite result. Mandril consistently requires more bits than Lena because its stronger texture produces more significant high-frequency coefficients. Its PSNR also drops more sharply when Q is doubled.

3 Experiment 2: Zigzag and Raster Scans

The standard zigzag order was compared with row-major raster order. The decoder used the matching inverse order in each case, so coefficient positions after decoding were unchanged.

Image	Scan order	BPP	PSNR (dB)	MSE
Lena	**Zigzag**	**0.837616**	35.808515	17.069817
Lena	Raster	1.012146	35.808515	17.069817
Mandril	**Zigzag**	**1.454590**	34.203934	24.699432
Mandril	Raster	1.797150	34.203934	24.699432

Table 2: Rate and distortion for zigzag and raster scans.

PSNR and MSE are identical because scan order is a reversible permutation. Zigzag places low-frequency coefficients first and groups the likely high-frequency zeros near the end, allowing EOB to terminate each block earlier. Relative to raster order, zigzag reduces BPP by 17.24% for Lena and 19.06% for Mandril.

4 Experiment 3: Separable and Direct DCT

The standard transform computes $F = C(fC^T)$ in two 1D stages. The direct implementation computes every output coefficient with the required double summation:

$$F_{p,q} = \sum_{x=0}^{7} \sum_{y=0}^{7} C_{p,x} f_{x,y} C_{q,y}.$$

Only DCT computation was timed with `time.perf_counter()`. Each method received one warm-up call and was then measured three times over all 4096 blocks. The reported value is the median on an Apple M1 Pro using Python 3.12.13 and NumPy 2.5.4.

Image	Method	Median time (s)	BPP	Payload bits
Lena	**Two-stage 1D**	**0.008258**	0.837616	219544
Lena	Direct 2D	3.058908	0.837616	219544
Mandril	**Two-stage 1D**	**0.007563**	1.454590	381278
Mandril	Direct 2D	3.064902	1.454559	381270

Table 3: DCT-only timing and resulting bitstream rate.

The two-stage transform is approximately 370× faster for Lena and 405× faster for Mandril in this Python implementation. The first-block maximum coefficient difference is below 4×10^{-13} for both images, confirming that the transforms are numerically equivalent. Lena produces the same bitstream length. Mandril differs by only 8 payload bits and one file byte because the two arithmetic orders can place a coefficient on opposite sides of a rounding boundary; the bitrate is therefore effectively unchanged.

5 Conclusion

Quantization controls the main rate-distortion tradeoff: finer steps improve quality at a higher bitrate. Zigzag scanning changes rate but not reconstruction quality because it makes trailing-zero EOB coding more effective. Finally, DCT separability preserves the same transform result and bitrate while reducing computation time by roughly two orders of magnitude compared with explicit direct summation.